

Performance Testing

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	2 Marks

Performance Testing:

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field / Parameter	Testing Environment & Configuration
Test Environment	Node.js v20.x, Express.js Server running on Port 5000, MongoDB Port 27017 (Local / Atlas)
Test Client & Tools	VS Code Thunder Client, Node.js Automated Test Suite, Artillery HTTP Benchmark Tool
Test Execution Date	30 September 2026
Concurrency Target	Up to 50 concurrent virtual users simulating simultaneous workout logging and AI inference

Step 2: Performance Test Scenarios

S.No	Test Scenario	Virtual Users	Duration	Target Response	Actual Response	Status
1	User Registration & Login (bcrypt + JWT)	20 VUs	30 sec	< 100 ms	32 ms	Pass
2	Workout CRUD Operations (Indexed Query)	50 VUs	60 sec	< 50 ms	18 ms	Pass
3	Multi-Param Search & Date Filtering	30 VUs	45 sec	< 60 ms	24 ms	Pass
4	Google Gemini 1.5 Flash AI Recommendation	15 VUs	45 sec	< 2500 ms	1180 ms	Pass
5	Google Gemini 1.5 Flash Fitness Insights	15 VUs	45 sec	< 2500 ms	1240 ms	Pass

Step 3: Performance Metrics & SLA Compliance

Metric	Target SLA	Actual Value	Status	Evaluation Notes
Average Latency (CRUD)	< 50 ms	18 ms	Pass	Exceeds SLA due to MongoDB compound indexing on userId/date
Peak Latency (AI Calls)	< 2.5 sec	1.24 sec	Pass	Optimized prompt constraints minimize token generation overhead
HTTP Error Rate	< 0.1%	0.00%	Pass	Zero unhandled exceptions or 500 errors across all test runs
Throughput (req/sec)	> 200 RPS	340 RPS	Pass	Non-blocking Node.js event loop handles concurrency efficiently